

Exact Condition for a Tax Penalty on Risky Business Formation

Purpose.

This note gives an exact condition, not a Taylor approximation, for when taxation makes a risky business less attractive than safe work after the business success payoff has been calibrated to pre-tax indifference.

The proof is deliberately written in terms of an arbitrary after-tax payoff mapping. This covers a simple realised capital-gain tax, a terminal fully franked dividend, and a corporate share sale with full or partial franking credit capitalisation.

Notation.

B background wealth, $B > 0$.

W pre-tax terminal accumulation from safe work.

W_T after-tax terminal accumulation from safe work.

p_f probability of business failure.

p_m probability of the same-as-safe business outcome.

p_h probability of high success, $p_h = 1 - p_f - p_m > 0$.

S high-success payoff, chosen by pre-tax indifference.

u increasing utility over total wealth; CRRA is specialised later.

$A(x)$ terminal after-tax business payoff when the pre-tax payoff is x .

For example, under a realised capital-gain tax, $A(x) = x - T(x)$. Under the imputation script, $A(x)$ is the state-specific after-tax amount generated by company tax, franking credits, dividend extraction, or share-sale treatment.

Assume the failure payoff receives no refund: $A(0) = 0$.

1. Pre-Tax Calibration

The risky business has pre-tax incremental payoffs

$X = 0$ with probability p_f ,

$X = W$ with probability p_m ,

$X = S$ with probability p_h .

The success payoff S is chosen so that the risky business has the same certainty equivalent as safe work before tax:

$$p_f u(B) + p_m u(B + W) + p_h u(B + S) = u(B + W). \quad (1)$$

Rearranging gives

$$p_h [u(B + S) - u(B + W)] = p_f [u(B + W) - u(B)]. \quad (2)$$

The same-as-safe probability p_m does not appear directly in (2), because that state gives exactly the benchmark payoff W . It still matters through $p_h = 1 - p_f - p_m$. If p_m rises while p_f is held fixed, p_h falls and the required success payoff generally rises.

Equation (2) is exact. It says that the probability-weighted utility gain from success exactly offsets the probability-weighted utility loss from failure.

2. Exact After-Tax Penalty Condition

The after-tax expected utility of the risky business is

$$EU_A = p_f u(B) + p_m u(B + A(W)) + p_h u(B + A(S)). \quad (3)$$

Let CE_A be its after-tax certainty equivalent:

$$u(B + CE_A) = EU_A.$$

The risky business is penalised relative to safe work exactly when

$$CE_A < W_T.$$

Because u is increasing, this is equivalent to

$$EU_A < u(B + W_T). \quad (4)$$

Now use the pre-tax indifference equation (1) to substitute for $p_f u(B)$:

$$p_f u(B) = u(B + W) - p_m u(B + W) - p_h u(B + S).$$

Substituting into (3) gives

$$\begin{aligned} EU_A &= u(B + W) \\ &- p_m [u(B + W) - u(B + A(W))] \\ &- p_h [u(B + S) - u(B + A(S))]. \end{aligned} \quad (5)$$

Combining (4) and (5), the exact necessary and sufficient condition for an after-tax penalty on the risky business is

$$\begin{aligned} &p_m [u(B + W) - u(B + A(W))] \\ &+ p_h [u(B + S) - u(B + A(S))] \\ &> u(B + W) - u(B + W_T). \end{aligned} \quad (6)$$

This is the exact condition. No Taylor expansion has been used.

3. Interpretation of the Exact Condition

Define the utility loss from taxing business payoff x as

$$L_A(x) = u(B + x) - u(B + A(x)).$$

Define the utility loss from taxing safe work as

$$L_{\text{safe}} = u(B + W) - u(B + W_T).$$

Then (6) can be written compactly as

$$p_m L_A(W) + p_h L_A(S) > L_{\text{safe}}. \quad (7)$$

This is exact and has a simple meaning:

The risky business is penalised when the probability-weighted utility loss from taxing the same and success states exceeds the utility loss from taxing the safe-work path.

Important implications.

1. The failure state drops out because $A(0) = 0$ and the same failure term appears before and after tax.

2. The same-as-safe state enters at the benchmark wealth level. It is not discounted by the marginal utility of a huge success payoff.

3. The success-state tax burden is exact utility loss $L_A(S)$. In a first order approximation it is downweighted by marginal utility at $B + S$, but condition (7) itself does not rely on that approximation.

4. The condition applies to imputation if $A(W)$ and $A(S)$ are interpreted as the after-tax terminal business payoffs produced by the imputation treatment in the code.

4. CRRA Specialisation

For CRRA utility, $\gamma > 0$:

$$u(w) = w^{(1 - \gamma)} / (1 - \gamma), \quad \gamma \neq 1,$$

$$u(w) = \log(w), \quad \gamma = 1.$$

The calibrated success payoff is exact.

For $\gamma \neq 1$, let $a = 1 - \gamma$. From (2):

$$(B + S)^a = (B + W)^a + (p_f / p_h) [(B + W)^a - B^a]. \quad (8)$$

If the right-hand side is in the range of w^a for $w > B + W$, then

$$S = \{ (B + W)^a + (p_f / p_h) [(B + W)^a - B^a] \}^{1/a} - B.$$

For log utility:

$$\log(B + S) = \log(B + W) + (p_f / p_h) [\log(B + W) - \log(B)], \quad (9)$$

so

$$S = \exp(\log(B + W) + (p_f / p_h) [\log(B + W) - \log(B)]) - B.$$

The exact penalty condition is obtained by substituting CRRA utility into

(6). For $\gamma \neq 1$:

$$\begin{aligned} & p_m * \{ (B+W)^a - (B+A(W))^a \} / a \\ & + p_h * \{ (B+S)^a - (B+A(S))^a \} / a \\ & > \{ (B+W)^a - (B+W_T)^a \} / a. \quad (10) \end{aligned}$$

For $\gamma = 1$:

5. Relation to the Approximate Condition

The earlier condition with marginal-utility weights is a first-order approximation to the exact condition.

Suppose $A(x) = x - T(x)$. For small tax wedges,

$$u(B + x) - u(B + x - T(x)) \approx u'(B + x) T(x).$$

Apply this to (6):

$$p_m u'(B+W) T(W) + p_h u'(B+S) T(S)$$

$$> u(B+W) - u(B+W-T).$$

Dividing by $u'(B+W)$ gives

$$p_m T(W) + p_h [u'(B+S)/u'(B+W)] T(S)$$

$$> [u(B+W) - u(B+W-T)] / u'(B+W). \quad (12)$$

For CRRA,

$$u'(B+S) / u'(B+W) = ((B+W)/(B+S))^\gamma.$$

So the approximate condition becomes

$$p_m T(W) + p_h ((B+W)/(B+S))^\gamma T(S)$$

$$> [u(B+W) - u(B+W-T)] / u'(B+W). \quad (13)$$

If the safe-work tax wedge is also linearised, the right-hand side is approximately $W - W-T$. This is useful intuition, but it is not the exact condition. The exact condition is (6), or (10)/(11) under CRRA.

6. What the Exact Proof Does and Does Not Imply

The exact proof supports the central intuition:

A progressive penalty can dominate deferral only if the utility loss from taxing the risky business states is large enough to exceed the utility loss from taxing the safe-work benchmark.

The proof also clarifies the role of the same-as-safe state:

With $p_m > 0$, the term $p_m L_A(W)$ enters exactly. This term is evaluated at the benchmark wealth level, so it can matter even when the success payoff S is so large that success-state tax dollars have low marginal utility.

The proof does not imply a universal gamma threshold. Statements such as 'gamma ≥ 1 means progressivity can never dominate deferral' require extra assumptions about the tax function, the safe-work tax path, the probability mix, and the limiting sequence being considered. The exact test is always condition (6).

In the R model, the practical implementation is:

1. Choose p_f , p_m , and $p_h = 1 - p_f - p_m$.
2. Solve S from the exact pre-tax indifference condition.
3. Compute $A(W)$ and $A(S)$ under the relevant tax treatment.
4. Compute the after-tax CE directly, or equivalently test (6).

This is why the three-state version is useful: it separates the tax burden on a moderate same-as-safe outcome from the tax burden on a rare extreme success outcome.